

01_Face_Detection(1)

November 7, 2025

```
[ ]:
```

0.1 Importing libraries and Loading dataset

```
[ ]: from PIL import Image
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import drive
drive.mount('/content/drive')
import zipfile
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
[ ]: path = '/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/09_Face_detection/
↳yalefaces.zip'
zip_object = zipfile.ZipFile(file = path, mode = 'r')
```

```
[ ]: zip_object.extractall('./')
zip_object.close()
```

```
[ ]: ls
```

```
drive/  __MACOSX/  sample_data/
yalefaces/
```

0.2 Preprocess the image:

```
[ ]: import os
print(os.listdir('/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/
↳09_Face_detection/yalefaces/yalefaces/train'))
```

```
['subject01.glasses.gif', 'subject01.noglasses.gif', 'subject01.leftlight.gif',
'subject01.normal.gif', 'subject01.sad.gif', 'subject01.rightlight.gif',
'subject01.surprised.gif', 'subject01.sleepy.gif', 'subject01.wink.gif',
'subject02.glasses.gif', 'subject02.happy.gif', 'subject02.noglasses.gif',
```

```

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'subject15.wink.gif', 'subject15.surprised.gif', 'subject04.sad.gif']

```

```

[ ]: def get_image_data():
    paths = [os.path.join('/content/drive/MyDrive/
↳MCV_Continuatin_inGoogle_Colab/09_Face_detection/yalefaces/yalefaces/
↳train',f) for f in os.listdir('/content/drive/MyDrive/
↳MCV_Continuatin_inGoogle_Colab/09_Face_detection/yalefaces/yalefaces/train')]

```

```

    #print(paths)

    faces = []
    ids = []

    for p in paths:
        #print(p)
        image = Image.open(p).convert('L') # Convert all the images into gray
        ↪scale.
        #print(image)
        #print(type(image)) # class

        image_np = np.array(image, 'uint8') # universal integer 8.
        #print(image_np)

        id = int(os.path.split(p)[1].split('.')[0].replace('subject',''))
        #print(id)

        ids.append(id)
        faces.append(image_np)

    return np.array(ids), faces

```

```
[ ]: ids, faces = get_image_data()
```

```
[ ]: np.unique(ids)
```

```
[ ]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15])
```

```
[ ]: ids
```

```
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```

```
[ ]: faces
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```

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```

```
[ ]: len(ids)
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```
[ ]: 135
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```
[ ]: len(faces)
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[ ]: 135
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[ ]: faces[100]
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```

```
[ ]: num_pixel = 243*320
```

```
[ ]: !pip install opencv-contrib-python==4.10.0.84
```

Requirement already satisfied: opencv-contrib-python==4.10.0.84 in
/usr/local/lib/python3.12/dist-packages (4.10.0.84)

Requirement already satisfied: numpy>=1.21.2 in /usr/local/lib/python3.12/dist-
packages (from opencv-contrib-python==4.10.0.84) (2.0.2)

0.3 Train the LBPH Classifier

```
[ ]: lbph_classifier = cv2.face.LBPHFaceRecognizer_create( radius = 4, neighbors = 14, grid_x = 9, grid_y = 9)
```

```
[ ]: #!pip install opencv-contrib-python==4.10.0.84
```

```
[ ]: lbph_classifier.train(faces, ids) #training
lbph_classifier.write('lbph_classifier.yml')
```

```
[ ]: ls
```

```
drive/ lbph_classifier.yml  __MACOSX/
sample_data/ yalefaces/
```

0.4 Recognizing faces:

```
[ ]: lbph_face_classifier = cv2.face.LBPHFaceRecognizer_create()
lbph_face_classifier.read('/content/lbph_classifier.yml')
```


0.5 Testing model from test image:

```
[ ]: # pass some image and check from test folder
sample_image = '/content/yalefaces/test/subject01.happy.gif'
image = Image.open(sample_image).convert('L')
np.array(image, 'uint8')

[ ]: array([[130, 130, 130, ..., 255, 255, 255],
           [255, 255, 255, ..., 255, 255, 255],
           [255, 255, 255, ..., 255, 255, 255],
           ...,
           [255, 255, 255, ..., 249, 255, 255],
           [255, 255, 255, ..., 255, 255, 255],
           [ 68,  68,  68, ...,  68,  68,  68]], dtype=uint8)

[ ]: image_np = np.array(image, 'uint8')

[ ]: # To predict
prediction = lbph_face_classifier.predict(image_np) # it gives label and
↳ confidence level.

[ ]: prediction

[ ]: (1, 63.26476332390736)

[ ]: expected_output = int(os.path.split(sample_image)[1].split('.')[0]
↳ replace('subject', ''))
expected_output

[ ]: 1

[ ]: cv2.putText(image_np, 'Pred: ' + str(prediction[0]), (10,30), cv2.
↳ FONT_HERSHEY_COMPLEX_SMALL, 1, (0,255,255))
cv2.putText(image_np, 'Expected: ' + str(expected_output), (10,60), cv2.
↳ FONT_HERSHEY_COMPLEX_SMALL, 1, (0,255,255))
cv2_imshow(image_np)
```

Pred: 1
Expected: 1



0.6 The above is for one image: ie; I gave one image from the test folder and predicted its output

```
[ ]:
```

0.7 Evaluate the face classifier I trained above:

```
[ ]: os.listdir('/content/yalefaces/test')
```

```
[ ]: ['subject10.centerlight.gif',  
      'subject14.sad.gif',  
      'subject12.rightlight.gif',  
      'subject12.normal.gif',  
      'subject13.sleepy.gif',  
      'subject11.glasses.gif',  
      'subject05.sleepy.gif',  
      'subject01.happy.gif',  
      'subject03.glasses.gif',  
      'subject15.rightlight.gif',  
      'subject06.leftlight.gif',  
      'subject02.leftlight.gif',  
      'subject04.leftlight.gif',  
      'subject09.rightlight.gif',  
      'subject09.sad.gif',  
      'subject14.normal.gif',
```

```
'subject02.centerlight.gif',
'subject10.sad.gif',
'subject01.gif',
'subject03.leftlight.gif',
'subject05.surprised.gif',
'subject08.rightlight.gif',
'subject11.happy.gif',
'subject13.sad.gif',
'subject06.happy.gif',
'subject15.sad.gif',
'subject08.normal.gif',
'subject07.leftlight.gif',
'subject07.happy.gif',
'subject04.surprised.gif']
```

```
[ ]: paths = [os.path.join('/content/yalefaces/test', f) for f in os.listdir('/content/yalefaces/test')] # get the path
```

```
[ ]: #paths
```

```
[ ]: predictions = []
expected_outputs = []

for p in paths:
    #print(p)
    image = Image.open(p).convert('L') # gray scale conversion
    image_np = np.array(image, 'uint8')

    prediction, _ = lbph_face_classifier.predict(image_np) # _--> ignore the
    #confidence level/ probability.

    expected_output = os.path.split(p)[1].split('.')[0].replace('subject', '') #
    #true label.

    predictions.append(prediction)
    expected_outputs.append(expected_output)
```

```
[ ]: # converting into array
predictions = np.array(predictions)
expected_outputs = np.array(expected_outputs, 'uint8')
```

```
[ ]: predictions
```

```
[ ]: array([ 4, 14, 12, 12, 13, 11,  5,  1,  3,  7, 14, 12,  4,  7,  9, 14,  4,
          10,  4,  3,  5,  4, 11, 13,  6, 15,  8,  9,  7,  4])
```

```
[ ]: expected_outputs
```

```
[ ]: array([10, 14, 12, 12, 13, 11,  5,  1,  3, 15,  6,  2,  4,  9,  9, 14,  2,
          10,  1,  3,  5,  8, 11, 13,  6, 15,  8,  7,  7,  4], dtype=uint8)
```

0.8 How accurate we are: Confusion Matrix

A confusion matrix is a table used to evaluate the performance of a classification model (especially supervised learning). It compares the actual (true) labels with the predicted labels from your model.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Example

Suppose: Actual = [1, 0, 1, 1, 0, 1, 0, 0] Predicted = [1, 0, 1, 0, 0, 1, 1, 0]

	Predicted 1	Predicted 0
Actual 1	3 (TP)	1 (FN)
Actual 0	1 (FP)	3 (TN)

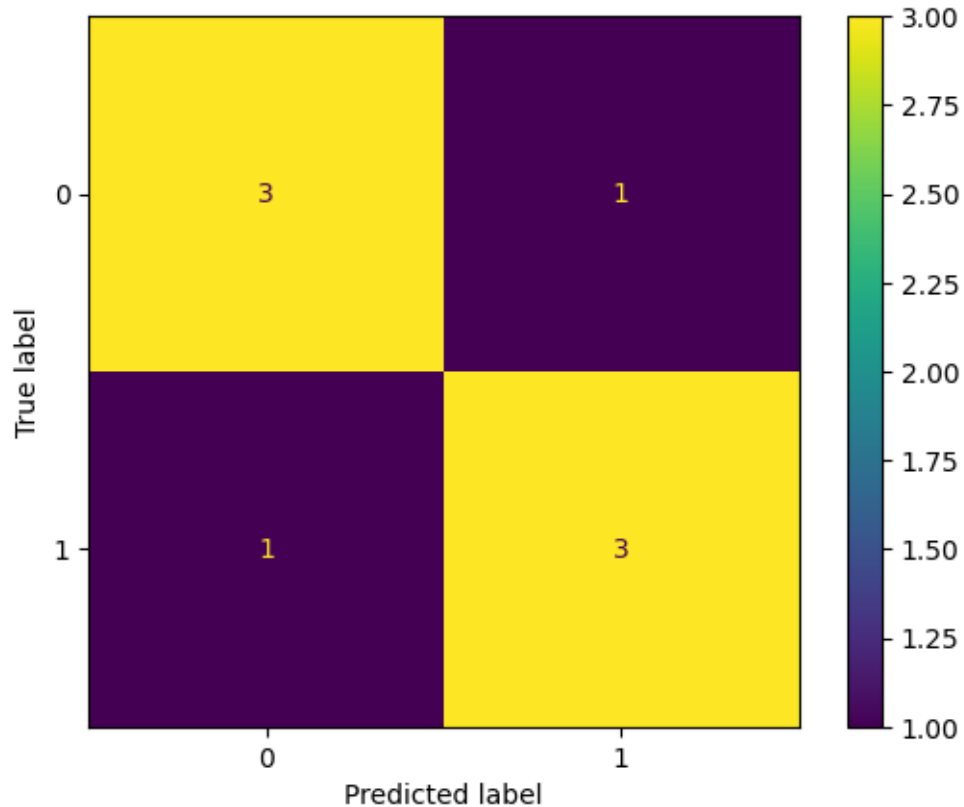
0.8.1 Example below:

```
[ ]: from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

y_true = [1, 0, 1, 1, 0, 1, 0, 0]
y_pred = [1, 0, 1, 0, 0, 1, 1, 0]

cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
```

```
[ ]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at
0x7e297bc826c0>
```



Accuracy: How often/ how many cases model is correct? $Acc = (TP+TN)/Total$ Accuracy paradox: Happens when we have an imbalanced data ? Accuracy

Definition: Measures how often the model correctly predicts the classes.

Accuracy= $\frac{TP+TN}{TP+TN+FP+FN}$ Accuracy= $\frac{TP+TN}{TP+TN+FP+FN}$

Interpretation: It tells you the proportion of total predictions that were correct.

Accuracy Paradox:

The accuracy paradox occurs when the dataset is imbalanced, i.e., one class is much more frequent than the other.

In such cases, a model can have high accuracy but still perform poorly on the minority class.

Example: If 95% of samples belong to Class A and only 5% to Class B, a model that always predicts Class A will have 95% accuracy, but it completely fails to detect Class B.

Metric	Formula	Meaning
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correctness
Precision	$TP / (TP + FP)$	How many predicted positives are correct

Metric	Formula	Meaning
Recall (Sensitivity)	$TP / (TP + FN)$	How many actual positives are captured
Specificity	$TN / (TN + FP)$	How many actual negatives are captured
F1-Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall

```
[ ]: from sklearn.metrics import accuracy_score, confusion_matrix
```

```
[ ]: accuracy_score(expected_outputs, predictions)
```

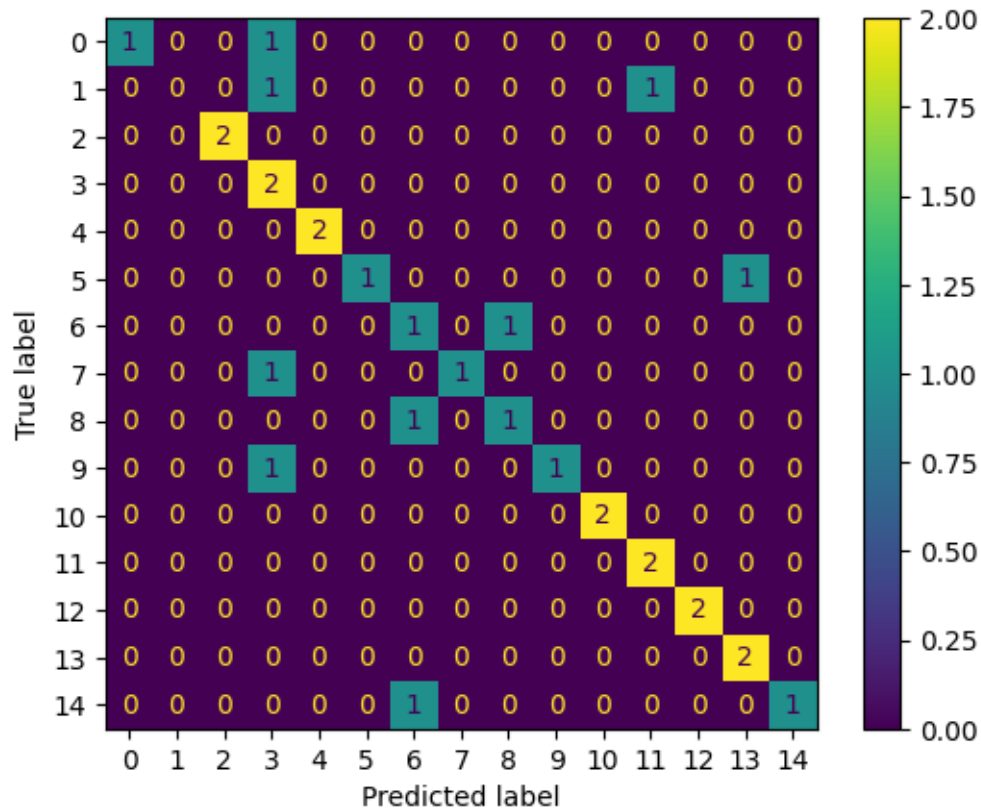
```
[ ]: 0.7
```

```
[ ]: cm = confusion_matrix(expected_outputs, predictions)
cm
```

```
[ ]: array([[1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
          [0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1],
          [0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0],
          [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0],
          [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0],
          [0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1]])
```

```
[ ]: disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
```

```
[ ]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at
0x7e2977d626f0>
```



```
[ ]: cm.shape
```

```
[ ]: (15, 15)
```

0.8.2 Other bunch of pre trained models:

<https://dlib.net/files/>

dlib is an open-source machine learning toolkit written in C++ with Python bindings. It's widely used for:

Face detection

Face landmark detection

Face recognition

Object detection

Image processing

Machine learning (SVM, kNN, regression, clustering, etc.)

```
[ ]:
```

0.9 Detecting Facial points:

- Using 68 points face detector.
- Using 5 points Face detector.

```
[ ]: import dlib

[ ]: face_detector = dlib.get_frontal_face_detector()
# Using 5 points face detector.
#points_detector = dlib.shape_predictor('/content/drive/MyDrive/
↳ MCV_Continuatin_inGoogle_Colab/09_Face_detection/
↳ shape_predictor_5_face_landmarks.dat')
# Using 68 points face detector.
points_detector = dlib.shape_predictor('/content/drive/MyDrive/
↳ MCV_Continuatin_inGoogle_Colab/09_Face_detection/
↳ shape_predictor_68_face_landmarks.dat/shape_predictor_68_face_landmarks.dat')

[ ]: image = cv2.imread('/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/
↳ 08_Object_Detection/people2.jpg')

[ ]: #image

[ ]: face_detection = face_detector(image,1)

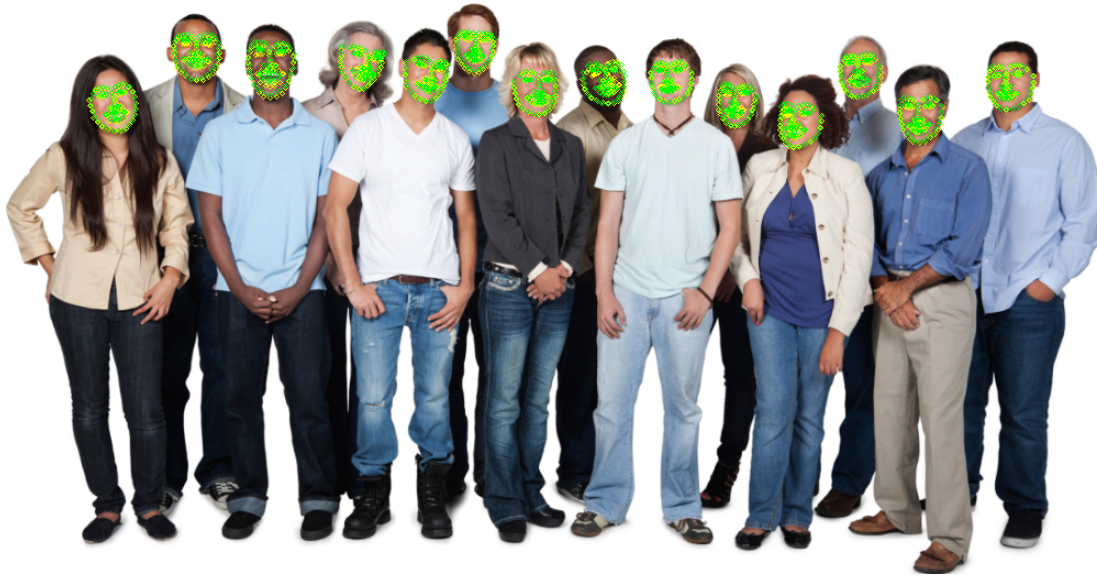
[ ]: face_detection

[ ]: rectangles[[ (429, 38) (465, 74)], [(665, 90) (701, 126)], [(717, 103) (760,
146)], [(909, 70) (952, 113)], [(828, 98) (871, 142)], [(605, 70) (641, 106)],
[(777, 62) (813, 98)], [(485, 78) (521, 114)], [(386, 60) (429, 103)], [(170,
41) (213, 84)], [(93, 89) (136, 132)], [(237, 50) (280, 94)], [(323, 50) (367,
94)], [(544, 65) (588, 108)]]

[ ]: len(face_detection)

[ ]: 14

[ ]: for face in face_detection:
    points= points_detector(image, face)
    #print(points.parts())
    for p in points.parts():
        cv2.circle(image, (p.x,p.y), 2, (0,255,0), 1)
cv2_imshow(image)
```

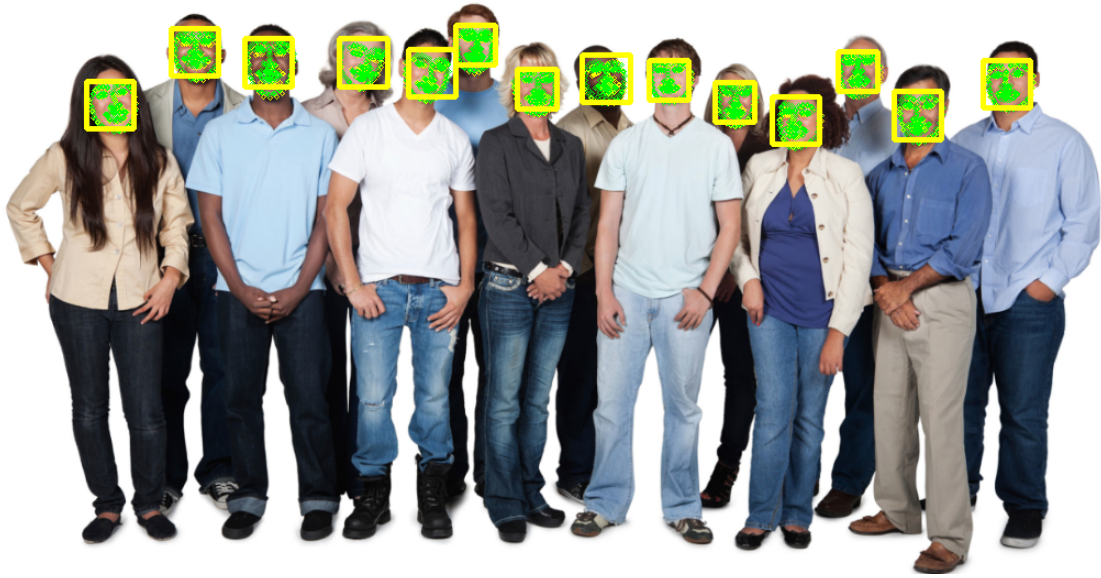



```
[ ]: face_detection # This gives rectangle.
```

```
[ ]: rectangles[[ (429, 38) (465, 74)], [(665, 90) (701, 126)], [(717, 103) (760, 146)], [(909, 70) (952, 113)], [(828, 98) (871, 142)], [(605, 70) (641, 106)], [(777, 62) (813, 98)], [(485, 78) (521, 114)], [(386, 60) (429, 103)], [(170, 41) (213, 84)], [(93, 89) (136, 132)], [(237, 50) (280, 94)], [(323, 50) (367, 94)], [(544, 65) (588, 108)]]
```

```
[ ]: for face in face_detection:
    points= points_detector(image, face)
    #print(points.parts())
    for p in points.parts():
        cv2.circle(image, (p.x,p.y), 2, (0,255,0), 1)
    l,t,r,b = face.left(), face.top(), face.right(), face.bottom()
    cv2.rectangle(image, (l,t),(r,b),(0,255,255),3)

cv2_imshow(image)
```



[]: